Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 1 of 10 · Question 10 part b · 4 marks

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10 (a)	For the correct value of $a = 10$	B1 [1]
(b)	Gradient of line L_2 $m = -\frac{1}{2}$ $y - '10' = '-\frac{1}{2}'(x-2)$ $y - 10 = -\frac{1}{2}(x-2)$ oe $x + 2y - 22 = 0$ *	B1 M1 A1 A1 cso [4]
(c)	Coordinates of point A are $(-3,\ 0)$ Coordinates of point B are $(22,\ 0)$ Length of AC $(5\sqrt{2})^2 = (m - '-3')^2 + n^2$ $[\Rightarrow 50 = m^2 + 6m + 9 + n^2]$ Gradient of BC $\frac{1}{4} = \frac{n}{m-'22'} \Rightarrow n = \frac{m-'22'}{4} \Rightarrow n^2 = \frac{m^2-44m+484}{16}$ $50 = m^2 + 6m + 9 + \frac{m^2-44m+484}{16}$ [or $50 = (22+4n)^2 + 6(22+4n) + 9 + n^2$] $17m^2 + 52m - 172 = 0$ OR $17n^2 + 200n + 575 = 0$ e.g. $m = \frac{-52 \pm \sqrt{52^2 - 4\times17\times-172}}{2\times17} \Rightarrow m = 2,\ (-5.058...)$ $m = 2$ and $n = -5$	B1 B1 M1 M1 ddM1 A1 M1 A1A1 [9]
(d)	Area$_{APB} = \frac{1}{2}('10') \times ('22' - '3') = (125)$ Area$_{ABC} = \frac{1}{2}('5') \times ('22' - '-3') = \left(\frac{125}{2}\right)$ Area of quadrilateral $ACBP$ $= \frac{375}{2}$ (units)^2	M1 A1 A1 [3]
	ALT	
	Area$_{ACBP} = \frac{1}{2}('25') \times ('15') = ...$ Area of quadrilateral $ACBP$ $= \frac{375}{2}$ (units)^2	[M1A1 A1]
	ALT	

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	$A=\frac{1}{2}\begin{bmatrix}'-3' & 2 & '22' & '2' & '-3'\\ 0 & '10' & 0 & '-5' & '0'\end{bmatrix}$	[M1
	$A=\frac{1}{2}\left(\left[(-3)\times 10+2\times 0+22\times(-5)+2\times 0\right]-\left[2\times 0+22\times 10+2\times 0+(-3)\times -5\right]\right)=\ldots$	A1
	Area of quadrilateral $APBC = \frac{375}{2}$ (units)^2	A1]
		Total 17 marks

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

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		Any other values of m and n must be rejected.
(d)	**M1**	For either area of triangle APB or ABC $\text{Area}_{APB} = \frac{1}{2}('10')\times('22'--'3') = (125)$ or $\text{Area}_{ABC} = \frac{1}{2}('5')\times('22'-'-3') = \left(\frac{125}{2}\right)$
	A1	Either area of triangle APB or ABC correct
	A1	Area of quadrilateral $ACBP = \frac{375}{2}$ (units)^2
ALT		
	M1A1	$\text{Area}_{ACBP} = \frac{1}{2}('25')\times('15') = \ldots$
	A1	Area of quadrilateral $ACBP = \frac{375}{2}$ (units)^2
ALT – determinant method		
	M1	For using the 'determinant method' e.g. $A = \frac{1}{2}\begin{bmatrix} '-3' & 2 & '22' & '2' & '-3' \\ 0 & '10' & 0 & '-5' & '0' \end{bmatrix}$
	A1	For a correct evaluation of their determinants using **their** values e.g. $A = \frac{1}{2}\left(\left[(-3)\times10+2\times0+22\times(-5)+2\times0\right]-\left[2\times0+22\times10+2\times0+(-3)\times-5\right]\right) = \ldots$
	A1	Area of quadrilateral $APBC = \frac{375}{2}$ (units)^2

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 1 of 10 · Question 10 part b · 4 marks

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Question	Scheme	Marks

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 2 of 10 · Question 2 part a · 4 marks

2023 JAN · Paper 01 · extract 1 of 5 · find part a in this mark scheme extract

2 a	(Gradient of AB =) $\frac{0-3}{4+5}\left(=-\frac{1}{3}\right)$	M1
	Gradient of the perpendicular = $-\frac{1}{"-\frac{1}{3}"}(=3)$	M1
	$y-5=-\frac{1}{"-\frac{1}{3}"}(x+1)$ or $y-5="3"(x+1)$ or $y=3x+8$	M1
	$3x-y+8=0$ or $-3x+y-8=0$ oe	A1 (4)
b	(Equation of AB =) $y-0="-\frac{1}{3}"(x-4)$ or $y=-\frac{1}{3}x+\frac{4}{3}$ oe	M1
	Solve $y-3x-8=0$ and $y-0=-\frac{1}{3}(x-4)$ oe simultaneously	M1
	$0=\frac{10}{3}x+\frac{20}{3}$ oe	
	$x=-2$ $y=2$ or (-2, 2) *	A1 cso (3)
	Alternative for final 2 marks (substitutes (- 2, 2) into each equation))	
	Line l is $3x-y+8=0$	
	When $x=-2$ and $y=2$, $-6-2+8=0\Rightarrow D$ lies on l	
	AB is $y=-\frac{1}{3}\ x-4$	{M1}
	When $x=-2$ and $y=2$, $-\frac{1}{3}\ -2-4\ =2=y\Rightarrow D$ lies on AB	
	So l and AB intersect at $-2, 2$ *	{A1 cso} (3)
c	Midpoint of $AB=\left(\frac{-5+4}{2},\frac{3(+0)}{2}\right)=\left(-\frac{1}{2},\frac{3}{2}\right)$	B1 (M1 on ePen)
	$\neq(-2,2)$ So not perpendicular bisector of AB *	B1 (A1 on ePen) cso (2)
ALT	$BD=\sqrt{(4--2)^2+(0-2)^2}\left(=\sqrt{40}\right)$ or $AD=\sqrt{(-5--2)^2+(3-2)^2}\left(=\sqrt{10}\right)$	M1
	$BD=\sqrt{40}(=2\sqrt{10})$ and $AD=\sqrt{10}$	A1cso
	$BD\neq AD$ so not the perpendicular bisector	(2)

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 2 of 10 · Question 2 part a · 4 marks

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d	$BD = \sqrt{(4--2)^2+(0-2)^2}\left(=\sqrt{40}\right)$ oe	M1
	$CD = \sqrt{(-1--2)^2+(5-2)^2}\left(=\sqrt{10}\right)$ oe	M1
	$\left(\tan\angle ABC=\right)\dfrac{"\sqrt{(-1--2)^2+(5-2)^2}"}{"\sqrt{(4--6)^2+(0-2)^2}"}$ or $\dfrac{\sqrt{10}}{\sqrt{40}}$	ddM1
	$=\dfrac{1}{2}$	A1 (4)
ALT 1	Uses cosine rule in triangle ABC	
	$AB = \sqrt{(-5-4)^2+(3-0)^2}\,(=\sqrt{90}=3\sqrt{10})$ or $BC = \sqrt{(4--1)^2+(0-5)^2}\,(=\sqrt{50}=5\sqrt{2})$	
	or $AC = \sqrt{(-1--5)^2+(5-3)^2}\,(=\sqrt{20}=2\sqrt{5})$	M1
	may also find expressions for $(AB)^2, (BC)^2$ or $(AC)^2$	
	$AB = \sqrt{(-5-4)^2+(3-0)^2}\,(=\sqrt{90}=3\sqrt{10})$ and $BC = \sqrt{(4--1)^2+(0-5)^2}\,(=\sqrt{50}=5\sqrt{2})$	
	and $AC = \sqrt{(-1--5)^2+(5-3)^2}\,(=\sqrt{20}=2\sqrt{5})$	M1
	may also find expressions for $(AB)^2, (BC)^2$ and $(AC)^2$	
	$\left(\cos ABC=\right)\dfrac{("\sqrt{50}")^2+("\sqrt{90}")^2-("\sqrt{20}")^2}{2\times"\sqrt{50}"\times"\sqrt{90}"}\left(=\dfrac{2}{\sqrt{5}}=\dfrac{2\sqrt{5}}{5}\right)$	ddM1
	$\left(\tan ABC\right)=\dfrac{1}{2}$	A1
		Total 13 marks

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 2 of 10 · Question 2 part a · 4 marks

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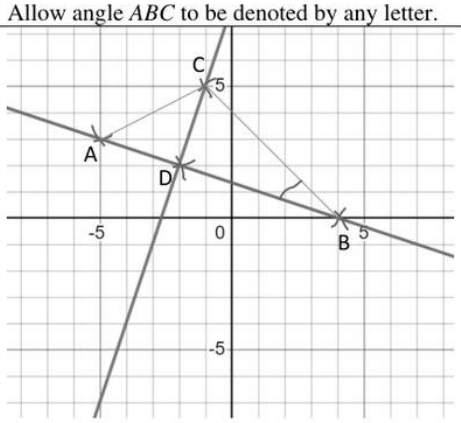
Part	Mark	Additional Guidance
(a)	M1	For correctly finding the gradient of AB, need not be simplified.
	M1	For finding the negative reciprocal of their gradient of AB, need not be simplified.
	M1	For a fully correct method using their negative reciprocal to find the equation of the straight line. Allow use of their $-\frac{1}{"-\frac{1}{3}"}$ which can be unsimplified or processed incorrectly, but must follow from a clear attempt to have found the negative reciprocal of their gradient for AB. If $y = mx + c$ is used, there must be a fully correct substitution and a fully correct rearrangement to find c
	A1	For $3x - y + 8 = 0$ or $-3x + y - 8 = 0$ oe so long as integer coefficients and = 0.
(b)	M1	For writing the equation of AB, the equation need not be simplified, but must be fully correct.
	M1	For an attempt to solve their equations simultaneously, allow one processing error. Must eliminate one variable and arrive at a value for x or y.
	A1 cso*	For obtaining the required result, no errors.
	ALT	**Final 2 marks**
	M1	For writing the equation of AB, the equation need not be simplified, but must be fully correct.
	M1	For a fully correct substitution of (–2, 2) into a correct equation for AB to show that D lies on AB **OR** a fully correct substitution of (–2, 2) into **their** equation for line l
	A1 cso*	For a fully correct substitution of (–2, 2) into a correct equation for AB to show that D lies on l **AND** a fully correct substitution of (–2, 2) into a correct equation for line l to show that D lies on l. An appropriate (minimal) conclusion must also be written, this can be as simple as # or 'shown'. No errors.
(c)	B1 (M1 on ePen)	For either coordinate of the midpoint correct
	B1 (A1 on ePen) cso*	For both coordinates correct and an appropriate (minimal) conclusion, which must be correct, can be as simple as # or 'shown'.
ALT	M1	For a correct method to calculate AD or BD
	A1 cso*	For correct values for AD and BD, accept decimals and a (minimal) conclusion, can be as simple as # or 'shown'.
(d)	M1	For the correct method to calculate BD (May be seen in (c))
	M1	For the correct method to calculate CD
	ddM1	For $\left(\tan\angle ABC =\right)\frac{\sqrt{(-1--2)^2+(5-2)^2}}{\sqrt{(4--6)^2+(0-2)^2}}$ or $\frac{\sqrt{10}}{\sqrt{40}}$ allow their BD and CD. Dependent on the previous 2 method marks.
	A1	For $\left(\tan\angle ABC\right)=\frac{1}{2}$
If you see a method using the gradient of BC *and* tan $(A - B)$ – PLEASE SEND TO REVIEW		

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Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 2 of 10 · Question 2 part a · 4 marks

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ALT 1	**M1**	Correct method to calculate one of the required lengths
	M1	Correct method to calculate all of the required lengths
	ddM1	Correct substitution into the cosine rule to give the expression shown, allow use of their lengths. Must obtain a correct expression for cosABC = , using their values. Dependent on the previous 2 method marks.
	A1	For $(\tan \angle ABC) = \frac{1}{2}$
Because of the wording of the question, in not specifically asking for an exact value, we are going to allow a range for the value of $\tan \angle ABC$ as follows: $0.49989 \leq \tan \angle ABC \leq \frac{1}{2}$		
If you see the candidate using the ratio for a right angled triangle - but they are using the wrong pairing of lengths, (perhaps AB, BC or AC) then this is to be marked using the main scheme, and is unlikely to be awarded any marks unless they have correctly calculated BD or CD. To be clear - If the candidate states correct values for AB, BC or AC but goes onto to use a tan ratio - this cannot be awarded any marks using the ALT mark scheme, because they are not using the correct cosine rule and with the tan ratio they should be using at least one of the correct sides, BD or CD. If you see a clear attempt to use the cosine rule, please use this ALT marking scheme. A special case is when the candidate has not made it clear which route they were intending to take - tan ratio (right angled triangle ratio) or cosine rule. If they have all lengths stated correctly, they could be awarded M1 M1 using either scheme.		
Further ALTS		If you see a valid method, leading to a fully correct exact value for tanABC, please award 4 marks. **If the value for tanABC is incorrect or approximate and there is some partially correct working, PLEASE SEND TO REVIEW.** Allow angle ABC to be denoted by any letter.
		 For reference $BD = \sqrt{40} (= 2\sqrt{10})$ $AD = \sqrt{10}$ $AB = \sqrt{90} = 3\sqrt{10}$ $BC = \sqrt{50} = 5\sqrt{2}$ $\sin ABC = \frac{\sqrt{5}}{5}$ $\cos ABC = \frac{2\sqrt{5}}{5}$

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 2 of 10 · Question 2 part a · 4 marks

2023 JAN · Paper 01 · extract 5 of 5 · find part a in this mark scheme extract

Question number	Scheme	Marks
	a	M1 A1

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 3 of 10 · Question 5 part c · 3 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

5(a)	$-\frac{1}{12}=\frac{10-0}{p-123}$ $\left[\text{or } -\frac{1}{12}=\frac{0-10}{123-p}\right]$ $\Rightarrow p-123=-12\times10\Rightarrow p=3*$	M1 A1 cso [2]
(b)	$y-10=-\frac{1}{12}(x-3)\Rightarrow 12y+x-123=0$	M1A1 [2]
(c)	$m_k=12$ $y-10=12(x-3)\Rightarrow y=12x-26$	B1 M1A1 [3]
(d)	At C: $y=12x-26$ when $y=0$, $x=\frac{26}{12}$ oe $\text{Area}_{ABC}=\frac{1}{2}\times10\times\left(123-\frac{13}{6}\right)=\frac{3625}{6}$ ALT $\text{Area}=\frac{1}{2}\begin{vmatrix}3 & 123 & 13/6 & 3\\ 10 & 0 & 0 & 10\end{vmatrix}=\frac{3625}{6}$	M1A1 M1A1 [M1A1] [4]
		Total 11 marks

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 3 of 10 · Question 5 part c · 3 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Question	Notes	Marks
5(a)	States a correct expression for the gradient in terms of p either $-\frac{1}{12}=\frac{10-0}{p-123}$ or $-\frac{1}{12}=\frac{0-10}{123-p}$ and attempts to solve their equation in p $p-123=-12\times10\Rightarrow p=...$	M1
	Finds the value of $p=3$* Must show an intermediate step e.g. $k=p\pm c$ or $c=\frac{p}{k}$ or $c=kp$	A1 cso [2]
(b)	Forms an equation using either the formula or $y=mx+c$ with the given values $y-10=-\frac{1}{12}(x-3)$ OR $y-0=-\frac{1}{12}(x-123)$	M1
	For the correct equation in the required form $12y+x-123=0$ Coefficients must be integers. Allow for $r=1, s=12, t=-123$	A1 [2]
(c)	For the gradient of the normal when $x=3$ is 12	B1
	Forms an equation of the normal using either the formula or $y=mx+c$ with their values of the gradient of the normal. $y-12=12(x-3)$	M1
	For the correct equation in the required form. $y=12x-26$	A1 [3]
(d)	Attempts to find the x coordinate at point C $y=12x-26$ when $y=0$, $x=...$ Their linear equation from (c) with $y=0$ and attempt to solve.	M1
	For the correct value of x $x=\frac{26}{12}$ oe	A1
	For attempting to find the area of triangle ABC Area $=\frac{1}{2}\times10\times\left(123-\frac{13}{6}\right)=...$ or Area $=\frac{1}{2}\begin{vmatrix}3 & 123 & 13/6 & 3\\10 & 0 & 0 & 10\end{vmatrix}$	M1
	For the correct area of the triangle $\frac{3625}{6}$ oe isw rounding	A1 [4]
		Total 11 marks

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 3 of 10 · Question 5 part c · 3 marks

2023 JAN · Paper 02 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 4 of 10 · Question 6 part b · 4 marks

2024 JUN · Paper 01 · extract 1 of 5 · find part b in this mark scheme extract

Question	Scheme	Marks
6 (a)	$p=1,\ \ q=8$	B1B1 [2]
(b)	$\left(\text{Gradient of } l=\dfrac{12-2}{3--2}=\right)2$	B1
	(Gradient of the perpendicular =) $-\dfrac{1}{"2"}$	B1ft
	Equation of k	
	$y-"8"="-\dfrac{1}{2}"(x-"1")$	M1
	$\Rightarrow 2y+x-17=0$ *	A1 cso [4]
(c)	When $y=0$, $(2\times 0+x-17=0\Rightarrow)\ x=17$	B1
	$(\lvert CD\rvert=)\sqrt{(0-"8")^2+("17"-"1")^2}=\sqrt{320}=\left[8\sqrt{5}\right]$	M1A1 [3]
(d)	(Length of CX =) $\dfrac{2\times 80}{8\sqrt{5}}\left(=4\sqrt{5}\right)$	B1
	$\left("4\sqrt{5}"\right)^2=(m-"1")^2+(n-"8")^2$	M1
	$"2"=\dfrac{n-"8"}{m-"1"}\Rightarrow[n=2m+6]$ oe **or** $y-2="2"(x+2)\Rightarrow[y=2x+6]$	M1
	or $y-12="2"(x-3)\Rightarrow[y=2x+6]$ oe	
	$\Rightarrow "80"=(m-"1")^2+(2m+6-"8")^2\ (\Rightarrow 0=5m^2-10m-75$ oe eg $0=m^2-2m-15)$	ddM1
	$\Rightarrow(5m+15)(m-5)=0$ oe eg $(m+3)(m-5)=0$	M1
	$m=5,(-3)$	A1
	$n(=2m+6=2\times 5+6)=16$	A1 [7]
ALT1	(Length of CX =) $\dfrac{2\times 80}{8\sqrt{5}}\left(=4\sqrt{5}\right)$	B1
	$(m-"17")^2+(n)^2=\left("4\sqrt{5}"\right)^2+\left("8\sqrt{5}"\right)^2$	M1
	$"2"=\dfrac{n-"8"}{m-"1"}\Rightarrow[n=2m+6]$ oe **or** $y-2="2"(x+2)\Rightarrow[y=2x+6]$	M1
	or $y-12="2"(x-3)\Rightarrow[y=2x+6]$ oe	
	$\Rightarrow "400"=(m-"17")^2+(2m+6)^2\ (\Rightarrow 0=5m^2-10m-75$ oe eg $0=m^2-2m-15)$	ddM1
	$\Rightarrow(5m+15)(m-5)=0$ oe eg $(m+3)(m-5)=0$	M1
	$m=5,(-3)$	A1
	$n(=2m+6=2\times 5+6)=16$	A1 [7]

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

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ALT2	(Length of CX =) $\frac{2\times 80}{8\sqrt{5}}\left(=4\sqrt{5}\right)$	B1
	Length of AB = $\sqrt{(12-2)^2+(3--2)^2}\left(=5\sqrt{5}\right)$	M1
	Vector $\overrightarrow{AB}=\begin{bmatrix}3--2\\12-2\end{bmatrix}\left(=5\begin{bmatrix}1\\2\end{bmatrix}\right)$	M1
	$\left(\left\|\overrightarrow{CX}\right\|=4\sqrt{5}\Rightarrow\right)\overrightarrow{CX}=4\begin{bmatrix}1\\2\end{bmatrix}=\begin{bmatrix}4\\8\end{bmatrix}$	ddM1
	$\Rightarrow$ Coordinates of X are $(1,8)+(4,8)=(5,16)$	M1A1
	$(m=5 \quad n=16)$	A1 [7]
ALT3	$\frac{1}{2}\begin{vmatrix}17 & m & 1 & 17\\0 & n & 8 & 0\end{vmatrix}$	B1
	$\frac{1}{2}\begin{vmatrix}"17" & m & "1" & "17"\\0 & n & "8" & 0\end{vmatrix}=80\Rightarrow\frac{1}{2}\left(17n+"8"m-("136"+n)\right)=80\left(\Rightarrow 2n+m=37\right)$	M1
	$"2"=\frac{n-8}{m-1}\Rightarrow[n=2m+6]$ oe **or** $y-2="2"(x+2)\Rightarrow[y=2x+6]$ oe **or** $y-12="2"(x-3)\Rightarrow[y=2x+6]$ oe	M1
	$\frac{1}{2}\begin{vmatrix}"17" & m & "1" & "17"\\0 & "2m+6" & "8" & 0\end{vmatrix}$ or $\frac{1}{2}\left(17("2m+6")+"8"m-("136"+"2m+6")\right)=80$	ddM1
	or $2("2m+6")+m=37\Rightarrow m=$	M1
	$m=5,\ n=16$	A1A1 [7]
		Total 16 marks

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 4 of 10 · Question 6 part b · 4 marks

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Part	Mark	Notes
(a)	B1	For $p = 1$ OR $q = 8$ Allow anything that clearly implies these values eg (1, 8) or 1, 8
	B1	For $p = 1$ AND $q = 8$ Allow anything that clearly implies these values eg (1, 8) or 1, 8
(b)	B1	For the gradient of $l = 2$
	B1ft	For the gradient of $k = -\frac{1}{"2"}$. Ft their calculation of the gradient of l.
	M1	For the correct unsimplified equation for k, using their p and q and any **changed** gradient. If the candidate uses $y = mx + c$, a fully correct rearrangement to find c must be shown (this can be implied by a correct c for their equation) and concluded with the equation of the line written.
	A1 cso	For the equation of k in the required form. Minimum steps shown, no errors or omissions. This must from a correct p and q
Watch for methods using vectors. If the candidate arrives at the correct unsimplified equation with no obviously incorrect work, following vector work, allow B1 B1ft M1 and the final A1 if the equation is given in the correct form. (See example in Practice)		
(c)	B1	For the x coordinate of $D = 17$
	M1	For a correct **method** to find the length of CD using their p and q and their 17
	A1	For the correct length of $\sqrt{320}$ oe
(d)	B1	Any correct unsimplified calculation for CX
	M1	For correctly using Pythagoras to form an equation in terms of m and n or x and y. Follow through their CX (ie coming from correct calculation with their p, q, their point D).
	M1	For an unsimplified equation in terms of m and n or x and y correctly using the gradient of l. Follow through their gradient for l, their p and their q. Allow working in p, q or x, y Note this work is sometimes being seen in (b). **As long as it is used in part (d),** the equation found in (b) may be given credit.
	ddM1	For eliminating either m or n from either equation and forming a 3TQ. Allow errors in processing if an initial correct method following from 2 correct unsimplified equations to eliminate one of the variables is seen. Allow working in p, q or x, y Dependent on both previous method marks. Note, the previous M mark is for the unsimplified equation, so if simplified incorrectly, this can be used here to gain this mark (general principle of marking).
	M1	For a minimally acceptable attempt to solve (see general guidance) their 3TQ This mark can be implied from a correct value of m = 5. If the quadratic is not the correct quadratic, method must be shown.
	A1	For one of $m = 5$ or $n = 16$
	A1	For both $n = 16$ and $m = 5$
ALT1	First B, final 2 A marks – as main scheme	
	M1	For correctly using Pythagoras to form an equation in terms of m and n or x and y. Follow through their CX and CD (ie coming from correct calculation with their p, q, their point D).
	M1	For an unsimplified equation in terms of m and n or x and y correctly using the gradient of l. Follow through their gradient for l, their p and their q. Allow working in p, q or x, y Note this work is sometimes being seen in (b). **As long as it is used in part (d),** the equation found in (b) may be given credit.
	ddM1	For eliminating either m or n from either equation and forming a 3TQ. Allow errors in processing if an initial correct method following from 2 correct unsimplified equations to eliminate one of the variables is seen. Allow working in p, q or x, y Dependent on both previous method marks. Note, the previous M mark is for the unsimplified equation, so if simplified incorrectly, this can be used here to gain this mark (general principle of marking).
	M1	For a minimally acceptable attempt to solve (see general guidance) their 3TQ This mark can be implied from a correct value of m = 5. If the quadratic is not the correct quadratic, method must be shown.

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ALT2	B1	Any correct unsimplified calculation for CX
	M1	Correct method to find the length of AB
	M1	Correct method to find vector AB
	ddM1	Correctly deduces the relationship between their vectors CX and AB Dependent on both previous method marks.
	M1	Correct method to find the coordinates of X using their vector work.
	A1	For one of $m = 5$ or $n = 16$
	A1	For both $n = 16$ and $m = 5$
ALT3	B1	For a fully correct array as shown oe
	M1	For placing their array (though ft their p and q and their "17") = 80 and correctly multiplying out the discriminant to form an unsimplified equation in m and n
	M1	For an unsimplified equation in terms of m and n or x and y correctly using the gradient of l. Follow through their gradient for l, their p and their q. Allow working in p, q or x, y Note this work is sometimes being seen in (a). **As long as it is used in part (d),** the equation found in (a) may be given credit.
	ddM1	For eliminating either m or n seen either in the array or in their equation formed. Allow errors in processing if an initial correct method following from 2 correct unsimplified equations to eliminate one of the variables is seen. Allow working in p, q or x, y Dependent on both previous method marks. Note, the previous M mark is for the unsimplified equation, so if simplified incorrectly, this can be used here to gain this mark (general principle of marking).
	M1	For solving their linear equation, allow one error in processing, leading to a value of m or n
	A1	For one of $m = 5$ or $n = 16$
	A1	For both $n = 16$ and $m = 5$
If a partial method is seen, with correct answers not from obvious incorrect working, marks may be given (general point of marking unless precluded from the mark scheme for a question). If a partial method is seen and marks cannot be fitted to the main or ALT schemes – send to review.		

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 4 of 10 · Question 6 part b · 4 marks

2024 JUN · Paper 01 · extract 5 of 5 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 5 of 10 · Question 6 part a · 5 marks

2020 JAN · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

Number		
6 (a)	$\frac{\text{Change in } y}{\text{Change in } x} = \frac{2-0}{2-3} = -2$	M1 A1
	$2 = 1 + c$	M1
	$c = 1$	A1 ft
	$x - 2y + 2 = 0$	A1
		(5)
(b)	$2y - 2 = 7y + 3$	M1
	$-5y = 5$	
	$y = -1$	A1
	When $y = -1$	
	$x = 2 \times -1 - 2 = -4$ So $C = (-4, -1)$	A1
	$\left(\frac{3-4}{2}, \frac{0-1}{2}\right) = \left(-\frac{1}{2}, -\frac{1}{2}\right)$	M1A1
		(5)
(c)	$AB = \sqrt{5}$	M1
	$BC = \sqrt{45}$	A1
	Area $= \frac{1}{2} \times \sqrt{5} \times \sqrt{45} \times \frac{1}{2}$	M1
	3.75	A1
		(4)
	Alternative c	**(14)**
	$\pm\frac{1}{2}\begin{vmatrix} 3 & 0 & 1 \\ 2 & 2 & 1 \\ -4 & -1 & 1 \end{vmatrix}$	M1
	$\pm\frac{1}{2}\left[3\begin{vmatrix} 2 & 1 \\ -1 & 1 \end{vmatrix} + 1\begin{vmatrix} 2 & 2 \\ -4 & -1 \end{vmatrix}\right]$	A1
	$\pm\frac{1}{2}[3(2+1) + (-2+8)] \times \frac{1}{2}$	M1
	3.75	A1

Part	Mark	Additional Guidance
(a)	M1	For an attempt to find the gradient using the given coordinates **and** a correct attempt to find the perpendicular gradient. Accept either $\frac{2-0}{2-3} = (-2)$ or $\frac{0-2}{3-2} = (-2) \Rightarrow m_p = -\frac{1}{'-2'}$
	A1	For $m = \frac{1}{2}$
	dM1	For a correct method to find the equation of a line $y - 2 = '\frac{1}{2}'(x-2)$ or $y - 0 = '\frac{1}{2}'(x-3)$ **The gradient must come from a correct attempt to find the gradient and the gradient of the perpendicular** If $y = mx + c$ is used, then they must use the correct values of x and y and a value for c must be reached before this mark is awarded.
	A1	For the correct equation in any form $y - 2 = \frac{1}{2}(x-2)$ or $y - 0 = \frac{1}{2}(x-3)$ or $y = \frac{1}{2}x + 1$ oe

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 5 of 10 · Question 6 part a · 5 marks

2020 JAN · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

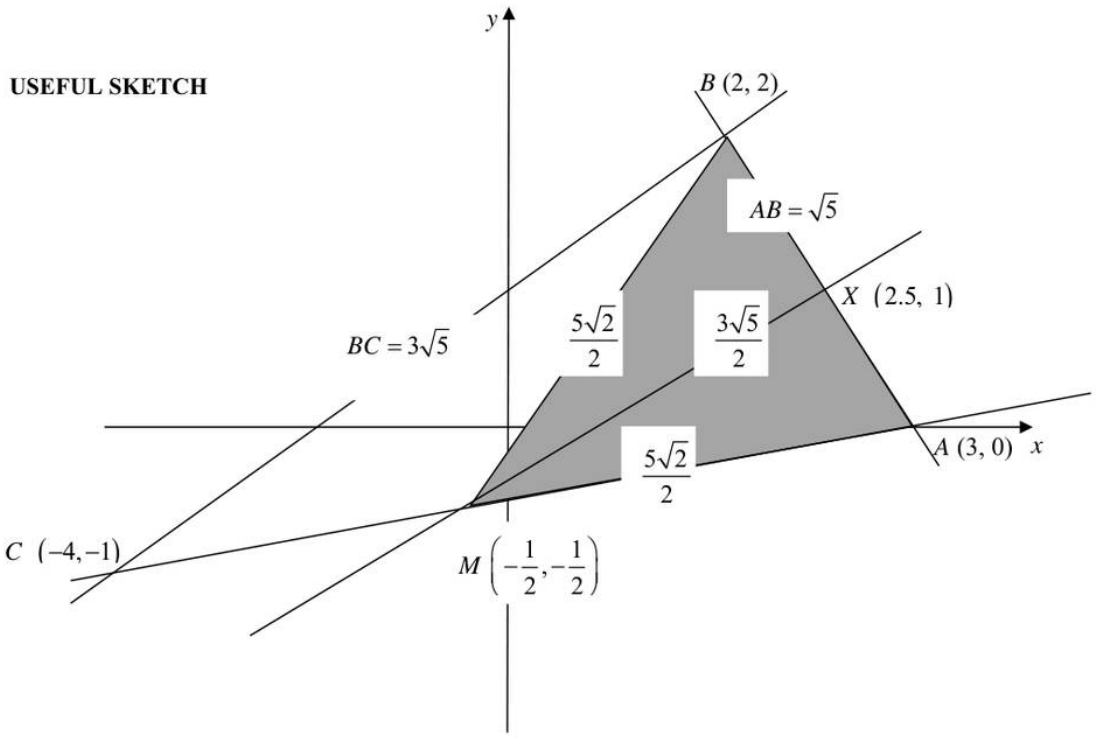
	A1	For the correct equation in the required form $x-2y+2=0$ **oe** arranged in any order but all one side (e.g. accept even $\frac{x}{2}-y+1=0$)	
(b)	M1	Sets $L_1=L_2$ **and** attempts to solve for y or x	
		$2y-2=7y+3$ $-5y=5\Rightarrow y=...$	$\frac{x+2}{2}=\frac{x-3}{7}\Rightarrow 7x+14=2x-6$ $5x=-20\Rightarrow x=...$
	A1	$y=-1$	$x=-4$
	A1	$x=-4$	$y=-1$
	M1	For any **correct** method to find the coords of M using **their** values for C of x and y and the given coordinates of A (3, 0) $\left(\frac{3+['-4']}{2},\frac{0+['-1']}{2}\right)$ **This is a B mark in Epen**	
	A1	$\left(-\frac{1}{2},-\frac{1}{2}\right)$ **This is a B mark in Epen**	
(c)	M1	For attempting to find the length AB **and** BC $AB=\sqrt{(3-2)^2+(0-2)^2}$ and $BC=\sqrt{(2--4)^2+(2--1)^2}$ **This is a B mark in Epen**	
	A1	For both $AB=\sqrt{5}$ and $BC=\sqrt{45}$ **This is a B mark in Epen**	
	M1	$\frac{1}{2}('\sqrt{5}'\times'\sqrt{45}')\times\frac{1}{2}$	For using a correct method to find the area of the triangle using correct lengths. i.e. they must be using BC and AB
	A1	For $A=3.75$	
	ALT – using determinants		
	M1	For using a correct method with their coordinates for C in any order (it is a triangle), but they must start and finish with the same coordinates $A=\frac{1}{2}\begin{pmatrix}3 & 2 & '-\frac{1}{2}' & 3\\ 0 & 2 & '-\frac{1}{2}' & 0\end{pmatrix}$ **This is a B mark in Epen**	
	A1	For using the correct coordinates $A=\frac{1}{2}\begin{pmatrix}3 & 2 & -\frac{1}{2} & 3\\ 0 & 2 & -\frac{1}{2} & 0\end{pmatrix}$ **This is a B mark in Epen**	
	M1	For a correct evaluation using their coordinates $A=\frac{1}{2}\left(\left[3\times2+2\times'-\frac{1}{2}'+'-\frac{1}{2}'\times0\right]-\left[2\times0+'-\frac{1}{2}'\times2+3\times'-\frac{1}{2}'\right]\right)=...$	

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 5 of 10 · Question 6 part a · 5 marks

2020 JAN · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

	A1	For $A = 3.75$
	ALT	
	M1	For finding the length $AB = \sqrt{(3-2)^2 + (0-2)^2}$ **and** $MX = \frac{1}{2}\sqrt{3^2 + \left(\frac{3}{2}\right)^2}$ (Let X be midpoint of AB so MX is height of triangle ABM)
	A1	$AB = \sqrt{5}$ $\quad MX = \frac{3\sqrt{5}}{2}$
	M1	Area of $\Delta ABM = \frac{1}{2} \times AB \times MX = \frac{1}{2} \times '\sqrt{5}' \times '\frac{3\sqrt{5}}{2}' = \left(\frac{15}{4}\right)$
	A1	Area of $\Delta ABM = \frac{15}{4} = 3.75$
If they use trigonometry, please send to review		



Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 5 of 10 · Question 6 part a · 5 marks

2020 JAN · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 6 of 10 · Question 3 part a · 5 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 4 · find part a in this mark scheme extract

number		
3 (a)	(Midpoint of AB =) $(5,5)$	B1
	(Gradient of AB =) $\frac{3-7}{9-1}=\left(-\frac{1}{2}\right)$ oe	M1
	(Gradient of l =) $-\left(\frac{1}{"-\frac{1}{2}"}\right)$ (= 2)	A1ft
	$y-"5"="2"(x-"5")$ oe	M1 A1 (5)
ALT	Let D (x, y) be any point on the line	
	$x-1^{\ 2}+\ y-7^{\ 2}=\ x-9^{\ 2}+\ y-3^{\ 2}$	{B1}
	$x^2-2x+1+y^2-14y+49=x^2-18x+81+y^2-6y+9$	{M1} {A1ft}
	$16x-40=8y \rightarrow (y=2x-5)$	M1 A1
(b)	$x=\frac{5}{2}$	B1ft
	(length of C to midpoint of AB =) $\sqrt{("5"-0)^2+("5"-2.5)^2}\left(=\frac{5\sqrt{5}}{2}\right)$	M1
	(length of AB =) $\sqrt{8^2+4^2}\left(=4\sqrt{5}\right)$	B1 (A1 on ePen)
	$\frac{1}{2}\times 4\sqrt{5}\times"\frac{5\sqrt{5}}{2}"$	M1
	25	A1 [5]
ALT1	Use of determinant	
	$x=\frac{5}{2}$	B1ft
	$\frac{1}{2}\begin{vmatrix}1 & 9 & "2.5" & 1\\ 7 & 3 & 0 & 7\end{vmatrix}$	M1 A1
	$\left(\frac{1}{2}\right)\left\|(1\times 3+9\times 0+"2.5"\times 7)-(1\times 0+"2.5"\times 3+9\times 7)\right\|$ oe	M1
	25	A1 [5]

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 6 of 10 · Question 3 part a · 5 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 4 · find part a in this mark scheme extract

ALT2	Area of rectangle – (sum of areas of 3 triangles)	
	$x = \frac{5}{2}$	B1ft
	$\frac{1}{2}\times\left("\frac{5}{2}"-1\right)\times 7+\frac{1}{2}\times\left(9-"\frac{5}{2}"\right)\times 3+\frac{1}{2}(9-1)(7-3)$ $\left(=\frac{21}{4}+\frac{39}{4}+16=31\right)$	M1
	$(9-1)\times 7=56$	A1
	$56-$ "31"	M1
	35	A1 [5]
		Total 10 marks

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 6 of 10 · Question 3 part a · 5 marks

2022 JAN · Paper 01 (regional R) · extract 3 of 4 · find part a in this mark scheme extract

Part	Marks	Notes
(a)	B1	For midpoint of $AB = (5,5)$
	M1	For correctly finding the unsimplified gradient of $AB = \frac{3-7}{9-1}$ oe
	A1ft	For correctly finding the unsimplified gradient of line l, use of their gradient for AB to find the negative reciprocal. $-\left(\frac{1}{"-\frac{1}{2}"}\right)$
	M1	For a correct method to find the equation of l using their midpoint and their gradient, which must have involved finding the negative reciprocal of the gradient of AB
	A1	For $y-5=2(x-5)$ ft oe
ALT	B1	For correctly equating the two expressions for AD^2 and AB^2 to give the equation shown
	M1	For an expansion of either their LHS or their RHS provided their equation is in the form. $x-a\ ^2+\ y-b\ ^2=\ x-c\ ^2+\ y-d\ ^2$ Allow one numerical or algebraic slip.
	A1ft	For a correct expansion of both their LHS and their RHS provided their equation is in the form. $x-a\ ^2+\ y-b\ ^2=\ x-c\ ^2+\ y-d\ ^2$
	M1	For simplifying leading to a linear equation
	A1	For $y=2x-5$ oe
(b)	B1ft	For $x=\frac{5}{2}$ ft their equation from part (a) with $y=0$
	M1	For finding the unsimplified length of C to midpoint of AB = $\sqrt{("5"-0)^2+("5"-"2.5")^2}\left(=\frac{5\sqrt{5}}{2}\right)$ Allow use of their (5, 5) and their C
	B1 (A1 on ePen)	For finding the unsimplified length of $AB=\sqrt{8^2+4^2}\left(=4\sqrt{5}\right)$
	M1	For area of triangle = $\frac{1}{2}\times 4\sqrt{5}\times"\frac{5\sqrt{5}}{2}"$
	A1	For 25
ALT1	B1ft	As main
	M1	For the determinant set up with their C
	A1	For the determinant set up fully correctly
	M1	For correctly evaluating their determinant, with their C, full working must be shown.
	A1	25
ALT2	B1ft	As main
	M1	For the correct calculation of the three areas using their C
	A1	For the correct rectangle area

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 6 of 10 · Question 3 part a · 5 marks

2022 JAN · Paper 01 (regional R) · extract 4 of 4 · find part a in this mark scheme extract

	M1	For the correct calculation, using their values for the three areas
	A1	25

Question number	Scheme	Marks

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 7 of 10 · Question 2 part a · 5 marks

2025 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

Number		
2 (a)	$(-3,9)$	B1
	$(m_{PQ}=)\dfrac{18-3}{3--7}\left(=\dfrac{3}{2}\right)$ oe	M1
	$(m_l=)-\dfrac{1}{\frac{18-3}{3--7}}=\left(-\dfrac{1}{\frac{3}{2}}\right)\left(=-\dfrac{2}{3}\right)$ oe	M1
	$y-"9"="-\dfrac{2}{3}"(x+"3")$	M1
	$3y+2x-21=0$ oe eg $-6y-4x+42=0$	A1 [5]
(b)	$3\times0+2x-21=0\Rightarrow x=(10.5)$	M1
	$(QR^2=)(3-"-3")^2+(18-"9")^2\left(=117\right)$ or $(QR=)\sqrt{(3-"-3")^2+(18-"9")^2}\left(=3\sqrt{13}\right)$	M1
	$(RS^2=)("10.5"-"-3")^2+(0-"9")^2\left(=\dfrac{1054}{4}\right)$ or $(RS=)\sqrt{("10.5"-"-3")^2+(0-"9")^2}\left(=\dfrac{9\sqrt{13}}{2}\right)$	M1
	$\dfrac{1}{2}\times"3\sqrt{13}"\times"\dfrac{9\sqrt{13}}{2}"$	ddM1
	$\dfrac{351}{4}$ oe (87.75)	A1 [5]
ALT1	$3\times0+2x-21=0\Rightarrow x=(10.5)$	M1
	$\left(\dfrac{1}{2}\right)\begin{vmatrix}3 & "-3" & "10.5" & 3\\ 18 & "9" & 0 & 18\end{vmatrix}$ oe eg $\left(\dfrac{1}{2}\right)\begin{vmatrix}"10.5" & "-3" & 3 & "10.5"\\ 0 & "9" & 18 & 0\end{vmatrix}$ oe	M1 M1
	$\dfrac{1}{2}\left((3\times9+-3\times0+10.5\times18)-(3\times0+10.5\times9+-3\times8)\right)$ oe	ddM1
	$\dfrac{351}{4}$oe (87.75)	A1
	Finding the area of a rectangle and subtracting area of triangles	
ALT2a	$3\times0+2x-21=0\Rightarrow x=0$	M1
	$("10.5"-"-3")\times18=243$	M1
	$\dfrac{1}{2}\times(18-"9")\times(3-"-3")(=27)$	M1 M1
	$\dfrac{1}{2}\times("10.5"-3)\times(18(-0))\left(=\dfrac{135}{2}\right)$	
	$\dfrac{1}{2}\times("10.5"-"-3")\times("9"(-0))\left(=\dfrac{243}{4}\right)$	
	$243-"27"-"\dfrac{135}{2}"-"\dfrac{243}{4}"$	ddM1
	$\dfrac{351}{4}$ oe (87.75)	A1

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 7 of 10 · Question 2 part a · 5 marks

2025 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

ALT2b	Let M be the intersection of PQ with the x-axis	M1
	$3\times0+2x-21=0\left(\Rightarrow x=10.5\right)$	M1
	$\left(y=\frac{3}{2}x+13.5\Rightarrow\ x=\right)-9$	M1
	$\left(\text{Area of triangle } SQM\ =\right)\frac{1}{2}\times\left("10.5"-"-9"\right)\times18\left(=\frac{351}{2}\right)$	M1
	$\left(\text{Area of triangle } SRM\ =\right)\frac{1}{2}\times\left("10.5"-"-9"\right)\times9\left(=\frac{351}{4}\right)$	ddM1
	$"\frac{351}{2}"-"\frac{351}{4}"$ $\frac{351}{4}$ oe (87.75)	A1
ALT3	Usually with a good diagram, the candidate can work out that triangles SRM and QRS are congruent (SAS) If spotting this, this is full marks or 0 marks (because without the correct coordinates, this is not so) $\frac{1}{2}\times19.5\times9=\frac{351}{4}$	M1M1 M1M1 ddM1A1
		Total 10 marks

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 8 of 10 · Question 8 part b · 5 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 5 · find part b in this mark scheme extract

Question	Scheme	Marks
8(a)	$\frac{y-5}{5-9}=\frac{x-1}{1-9}$ $\Rightarrow x-2y+9=0$	M1A1 A1 [3]
(b)	Coordinates of point X $\left(\frac{3\times9+1\times1}{3+1},\frac{3\times9+1\times5}{3+1}\right)=(7,8)$ The perpendicular gradient $=-\frac{1}{\frac{1}{2}}=-2$ Equation of l $y-8=-2(x-7)\Rightarrow y=-2x+22$ *	B1B1 B1ft M1A1 cso [5]
(c)	$y=-2(6)+22\Rightarrow p=10$	B1 [1]
(d)	$\overrightarrow{BA}=\begin{pmatrix}1-9\\5-9\end{pmatrix}=\begin{pmatrix}-8\\-4\end{pmatrix}\Rightarrow \overrightarrow{CD}=\begin{pmatrix}6-8\\10-4\end{pmatrix}=\begin{pmatrix}-2\\6\end{pmatrix}$ and coordinates of C are (6, 10) so coordinates of D are $(-2, 6)$	M1 A1A1 [3]
(e)	Length of AB (or CD) $=\sqrt{(9-5)^2+(9-1)^2}=\sqrt{80}$ Length of $CX=\sqrt{(10-8)^2+(7-6)^2}=\sqrt{5}$ Area of parallelogram $ABCD=\sqrt{5}\times\sqrt{80}=\sqrt{400}=20$ (units^2)	B1 B1 M1A1 [4]
		Total 16 marks

Part	Mark	Notes
(a)	M1	For using a correct method and the given coordinates of A and B to form the equation of AB. Do not score this mark until they find the gradient and form the equation of the line using a correct formula. If they use $y=mx+c$ do not allow this mark until they find c and form a complete equation. $\left[m=\frac{1}{2},\quad c=\frac{9}{2}\quad y=\frac{x}{2}+\frac{9}{2}\right]$
	A1	For the correct equation of AB in any form.
	A1	For the correct equation of AB in the required form.

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 8 of 10 · Question 8 part b · 5 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 5 · find part b in this mark scheme extract

(b)	B1	For either x or y correct coordinates of point X **NB This is an M mark in Epen**
	B1	For both x and y correct coordinates of point X **NB This is an A mark in Epen.**
	B1ft	For writing down the inverse reciprocal of their gradient for AB Ft their gradient form part (a)
	M1	For forming an equation for l using their coordinates of X and their negative reciprocal gradient of AB $y-8=-2(x-7)\Rightarrow y=-2x+22$
	A1	For the correct equation of l as shown.
(c)	B1	For $y = 10$
(d)	M1	For a suitable method. **Method 1 - Uses vectors:** $\overrightarrow{BA}=\begin{pmatrix}1-9\\5-9\end{pmatrix}=\begin{pmatrix}-8\\-4\end{pmatrix}\Rightarrow \overrightarrow{CD}=\begin{pmatrix}6+(-8)\\10+(-4)\end{pmatrix}\Rightarrow\left[\begin{pmatrix}-2\\6\end{pmatrix}\right]$ OR: $\overrightarrow{BC}=\begin{pmatrix}6-9\\9-10\end{pmatrix}=\begin{pmatrix}-3\\-1\end{pmatrix}\Rightarrow \overrightarrow{AD}=\begin{pmatrix}1-3\\5-(-1)\end{pmatrix}\Rightarrow\left[\begin{pmatrix}-2\\6\end{pmatrix}\right]$ **Method 2 - Uses simultaneous equations**: The equation of AD is $y=-\frac{x}{3}+\frac{16}{3}$ and of CD is $y=\frac{x}{2}+7$ $-\frac{x}{3}+\frac{16}{3}=\frac{x}{2}+7\Rightarrow x=-2$ and $y=6$ **Method 3 - Uses the gradient and length of *AB* or *CD*** $AB=CD=4\sqrt{5}=\sqrt{(6-x)^2+(y-10)^2}$ Gradient: $\frac{1}{2}=\frac{10-y}{6-x}\Rightarrow x=2y-14$ $\left(4\sqrt{5}\right)^2=(6-(2y-14))x^2+(y-10)^2\Rightarrow 5y^2-100y+420=0$ $\Rightarrow y=6,14$ and $x=-2,14$ Allow no more than one error in either method.
	A1	For either correct x or y coordinate of D [(– 2, 6)] **NB** this is an A mark in Epen
	A1	For both correct coordinates of D (– 2, 6)

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 8 of 10 · Question 8 part b · 5 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 5 · find part b in this mark scheme extract

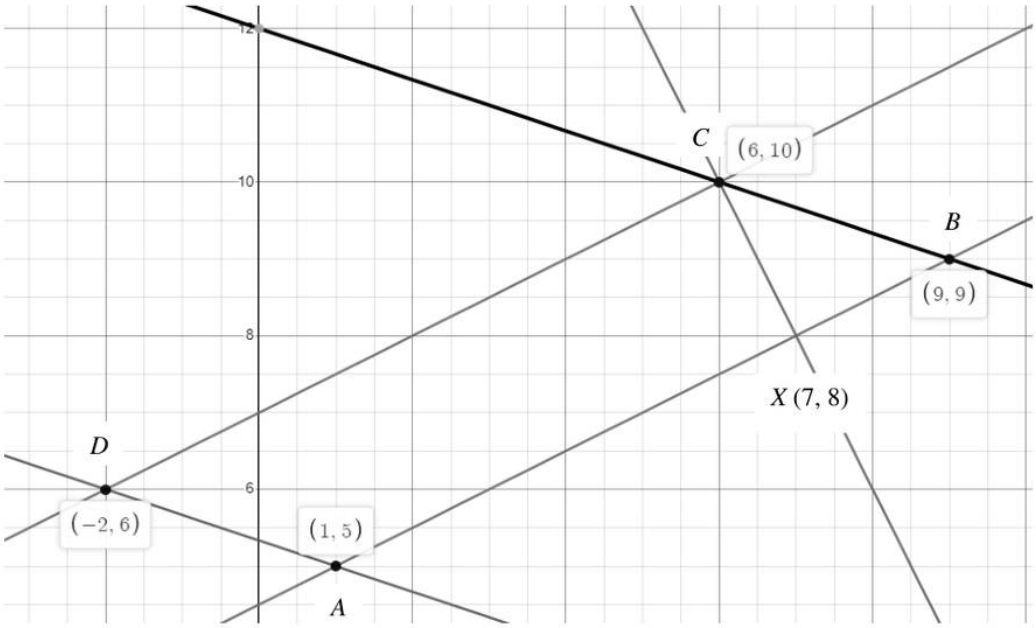
(e)	B1	For the correct length of either ***AB*** **[*CD*] or *CX*** These are given coordinates there is no ft
	B1	For the correct lengths of both ***AB*** **[*CD*] and *CX*** Ft their C and their X
	M1	For using a correct method to calculate the area of a parallelogram. $b \times h = CX \times AB = \sqrt{5} \times \sqrt{80} = \sqrt{400} = 20$ If their lengths are incorrect allow this mark provided their lengths are identified as base and perpendicular height. If they use AD as the base $\left[\sqrt{10}\right]$, the perpendicular height required is $2\sqrt{10}$ so base × height = $\sqrt{10} \times 2\sqrt{10} = ...$
	A1	For the correct area of 20 [units^2]
	ALT – Using determinants	
	B1B1	For sight of the correct array which must use the coordinates of A, B, C and D **ONLY.** The coordinates (7, 8) seen in the array is B0B0 $\begin{pmatrix} 1 & 9 & '6' & '-2' & 1 \\ 5 & 9 & '10' & '6' & 5 \end{pmatrix}$ Award both marks for fully correct. Award B1 if there are no more than 2 errors but with none missing. There must be 5 sets of coordinates in the array with first and last the same. **The coordinates must go in order around the parallelogram clockwise or anticlockwise.**
	M1	For the correct evaluation of their 2×5 array. Allow this even if they have (7, 8) included instead of (9, 9) [which is a common error] $\frac{1}{2}\begin{pmatrix} 1 & 9 & 6 & -2 & 1 \\ 5 & 9 & 10 & 6 & 5 \end{pmatrix} = \frac{1}{2}\left\lvert\left[(9+90+36-10)-(45+54-20+6)\right]\right\rvert = ...$
	A1	For the correct area of 20 [units^2]

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 8 of 10 · Question 8 part b · 5 marks

2023 JUN · Paper 02 (regional R) · extract 4 of 5 · find part b in this mark scheme extract

Useful sketch



Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 8 of 10 · Question 8 part b · 5 marks

2023 JUN · Paper 02 (regional R) · extract 5 of 5 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 9 of 10 · Question 4 part b · 6 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Question	Scheme	Marks
4(a)	$(3\times14-2\times12-p=0\Rightarrow)\ p=18$	B1
	$3y-2q-"18"=0\Rightarrow 3\times2-2q-"18"=0\Rightarrow q=$ $q=-6$	M1 A1 [3]
(b)	$(X(x,y)=)\left(\frac{1\times"-6"+2\times12}{3},\frac{1\times2+2\times14}{3}\right)$ oe $=(6,10)$	B1B1 or M1A1
	$(\text{Gradient of } AB=)=\frac{2}{3}$ oe	B1
	$(\text{Gradient of } L=)-\frac{1}{"\frac{2}{3}"}\left(=-\frac{3}{2}\right)$	B1ft
	$y-"10"=-\frac{1}{"\frac{2}{3}"}(x-"6")\left(\Rightarrow y=-\frac{3}{2}x+19\right)$	M1
	$\Rightarrow 3x+2y-38=0$ oe Where a, b and c are integers.	A1 [6]
		Total 9 marks

	Mark	Notes
(a)	B1	For $p = 18$
	M1	Uses their value of p and substitutes $y = 2$ and finds a value for q.
	A1	For the correct value of q.
First 2 marks of (b) – use the scheme which gives the most marks to candidates.		
(b)	B1 (M1 ePen)	For one correct value (6, 10).
	B1 (A1 ePen)	For both correct values (6, 10).
	or M1	Correct method for finding both coordinates, using their q. Look for any equivalent methods eg similar triangles.
	or A1	Both correct values.
	B1	For a fully correct unsimplified or simplified gradient of AB Note this mark is awarded for **finding or stating** the gradient and if simplified, it must be clear that $\frac{2}{3}$ is not a fraction from the given ratio. Accept $\frac{14-2}{12--6}$ This mark may be implied for a correct gradient of $-\frac{3}{2}$ used in further work.
	B1ft	For finding the negative reciprocal of their gradient – does not need to be simplified.
	M1	For a full and correct method to find the equation for the line L using **their** coordinates of X and **their** perpendicular gradient. This need not be simplified. If $y = mx + c$ is used, the correct value of c must be found for their coordinates and gradient. It must be clear the candidate is using their coordinates for X and not, for example simply the numbers from the coordinates of A and B
	A1	For the correct equation of the line in the required form – any equivalent where a, b and c are integers.

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 9 of 10 · Question 4 part b · 6 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 10 of 10 · Question 11 part a · 6 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 5 · find part a in this mark scheme extract

11 (a)	$b = 33$	B1
	$\left(\left(\frac{3\times0+2\times10}{5}, \frac{3\times3+2\times33}{5}\right)\right) = (4, 15)$	B1 B1
	ALT	
	Identifies the correct x coordinate of $x = 4$ using any method.	B1
	$y = 3\times'4'+3 = ... \Rightarrow y = 15$	B1B1
	For the gradient of line k $\quad m = -\frac{1}{"3"}$	M1
	$y - "15" = -\frac{1}{"3"}(x - "4")$	dM1
	$3y + x - 49 = 0^*$ cso	A1* cso [6]
(b)	$A = (0, 3)$	B1
	$-\frac{1}{"3"} = \frac{q-"3"}{p(-0)} (\Rightarrow p = -"3"(q-"3"))$ **alt** $y = -\frac{1}{3}x + 3 \Rightarrow q = -\frac{p}{3} + 3$ **OR** $p^2 + (q-"3")^2 = 1440$	M1
	Equation in *q* $10q^2 - 60q + 90 = 1440 \Rightarrow q^2 - 6q - 135 = 0$ oe $10(q-3)^2 = 1440$ or $q^2 - 6q - 135 = 0$	dM1
	Equation in *p* $1440 = p^2 + \left(3 - 3 - \frac{p}{3}\right)^2 \Rightarrow 1440 = \frac{10p^2}{9}$ And attempting to solve by any valid method, reaching a value for p	
	$q - 3 = \pm\sqrt{144}$ or $(q-15)(q+9)$ or $\frac{--6 \pm \sqrt{(-6)^2 - 4.(1).(-135)}}{2\times1}$	ddM1
	$q = 15, p = -36$	A1 A1 [6]
	ALT	
	$A = (0, 3)$	[B1
	$a\sqrt{(-3)^2 + 1^2}$ $(= 12\sqrt{10})$	M1
	$\Rightarrow a = (12)$	dM1
	$p = 0 - 3\times"12"$	dM1
	$q = 3 + 1\times"12"$	
	$p = -36$ and $q = 15$ or D is $(-36, 15)$	A1 A1]

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 10 of 10 · Question 11 part a · 6 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 5 · find part a in this mark scheme extract

(c)	**Finding the coordinates of points *E* and *F***	
	(Finding E $x = -36$ $3y + x - 49 = 0$)	M1
	$3y + ("-36") - 49 = 0 \left(y = \frac{85}{3} \right)$	
	(Finding F $x = -36$ $y - 3x = 3$)	M1
	$y - 3("-36") = 3$ $(y = -105)$	
	Find the area of triangle *ECF*	
	Method 1	
	$(CE =) \sqrt{("4" - "-36")^2 + \left("\frac{85}{3}" - "15"\right)^2} = \left(\frac{40\sqrt{10}}{3}\right)$	M1
	Or	
	$(CF =) \sqrt{("4" - "-36")^2 + ("15" - "-105")^2} = \left(40\sqrt{10}\right)$	
	$\left(\text{Area triangle } ECF = \right) \frac{1}{2} \times \frac{40\sqrt{10}}{3} \times 40\sqrt{10} = \frac{8000}{3}$ oe	M1A1 [5]
	Method 2	
	$\text{Area} = \frac{1}{2} \begin{bmatrix} "-36" & "4" & "-36" & "-36" \\ "\frac{85}{3}" & "15" & "-105" & "\frac{85}{3}" \end{bmatrix}$ oe	M1
	$\frac{1}{2} \left(\begin{bmatrix} "-36" \times "15" + "4" \times "-105" + "-36" \times "\frac{85}{3}" \\ - \left["-36" \times "-105" + "-36" \times "15" + + "4" \times "\frac{85}{3}" \right] \end{bmatrix} \right) = \left(\frac{10060}{3} - [-1980] \right) = \frac{8000}{3}$ oe	M1A1
	Method 3	
	Finds the lengths: $DC = 4 - (-36) = 40$ and $EF = \frac{85}{3} - (-105) = \frac{400}{3}$	M1
	$\text{Area} = \frac{1}{2} \times '40' \times '\frac{400}{3}' = \frac{8000}{3}$ oe	M1A1
		Total 17 marks

Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 10 of 10 · Question 11 part a · 6 marks

2022 JUN · Paper 01 (regional R) · extract 3 of 5 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1	Correct value of b. This may be embedded in a coordinate of B (10, 33)
	B1	For either $x = 4$ or $y = 15$
	B1	For (4, 15) or both $x = 4$ and $y = 15$
	ALT (Without finding the value of b)	
	B1	For $x = 4$ [by any method]
	B1	Substitutes their value of $x = 4$ into the given equation of l $y = 3\times\text{'4'}+3 = ...$
	B1	For $y = 15$
	M1	For identification of a gradient from the equation of l and use of $m = -\frac{1}{\text{"3"}}$
	dM1	For the correct use of any form of the equation of a line, if using $y = mx + c$, must reach $c = ...$ Note that $c = \frac{49}{3}$ Use of their (4, 15) which **must** have come from an attempt to find the midpoint of AB and their m. **Note:** Dependent upon the previous method mark.
	A1	$3y + x - 49 = 0$ accept terms on one side in any order. Cso **Note:** This equation is given. Award this and the previous mark that has come from work without any errors or omissions.
(b)	B1	Identifies the y intercept of line l (0, 3). This can be implied by later correct work or accept seen in part (a) Look for this coordinate written on a sketch if it is not in the body of the work.
	M1	Equates $-\frac{1}{\text{"3"}}$ to a **correct** expression for the gradient in terms of p and q, using **their** values of (0, 3) for the y intercept. **OR** for **correct** use of Pythagoras for the length of AD to form an equation in p and q, using **their** value of (0, 3) for the y intercept.
	dM1	For forming a quadratic equation in either p or q which must come from using their expression for the gradient and the length. **This is dependent on the first M mark in (b)**
	ddM1	For solving their quadratic equation by any acceptable method. **This is dependent on the previous two M marks in (b)**
	A1	For a correct p **or** q.
	A1	For a correct p **and** q.
	ALT	
	B1	Identifies the y intercept of line l (0, 3) This can be implied by later correct work or accept seen in part (a) Look for this coordinate written on a sketch if it is not in the body of the work.
	M1	Uses a correct method, scaling the gradient of $-\frac{1}{3}$ Must arrive at a scaling value (shown as a in the mark scheme), but it may not be the correct value. This mark may be implicit in work that follows.
	dM1	Uses their scaling factor, with the coordinates of A, to find a value for p or q **This is dependent on the first M mark in (b)**
	dM1	Uses their scaling factor, with the coordinates of A, to find a value for p and q **This is dependent on the first M mark in (b)**
	A1 A1	First A1 for p **or** q Second A1 for p **and** q
(c)	M1	Substitutes their value for – 36 into the equation for line k to find the y coordinate of point E
	M1	Substitutes their value for – 36 into the equation for line l to find the y coordinate of point F

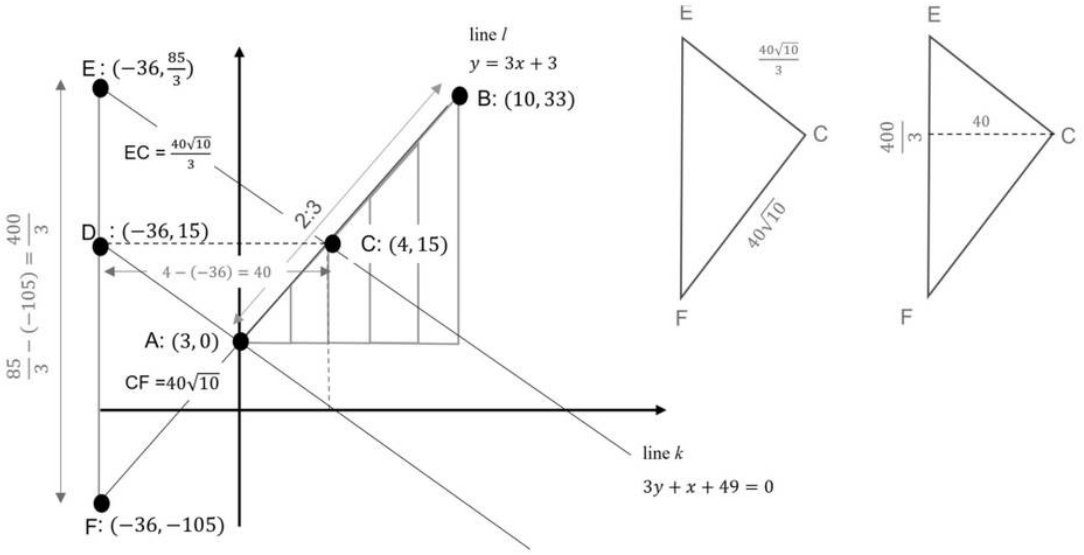
Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 10 of 10 · Question 11 part a · 6 marks

2022 JUN · Paper 01 (regional R) · extract 4 of 5 · find part a in this mark scheme extract

	Method 1	
	M1	Uses Pythagoras with **their** values to find the length of CE **and** CF correctly.
	M1	Correct method to find the area of the triangle, using **their** values.
	A1	For $\frac{8000}{3}$ oe
	Method 2	
	M1	For a correct array using their values.
	M1	Correct calculation shown, using their values, takes the modulus and multiplies by $\frac{1}{2}$
	A1	$\frac{8000}{3}$ oe
	Method 3	
	M1	For finding the lengths DC **and** EF correctly using **their** values.
	M1	Applies the correct formula for the area of a triangle using their values.
	A1	For $\frac{8000}{3}$ oe

USEFUL SKETCH



Type MS · Equation of a perpendicular line (including perpendicular bisectors)

Answer 10 of 10 · Question 11 part a · 6 marks

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